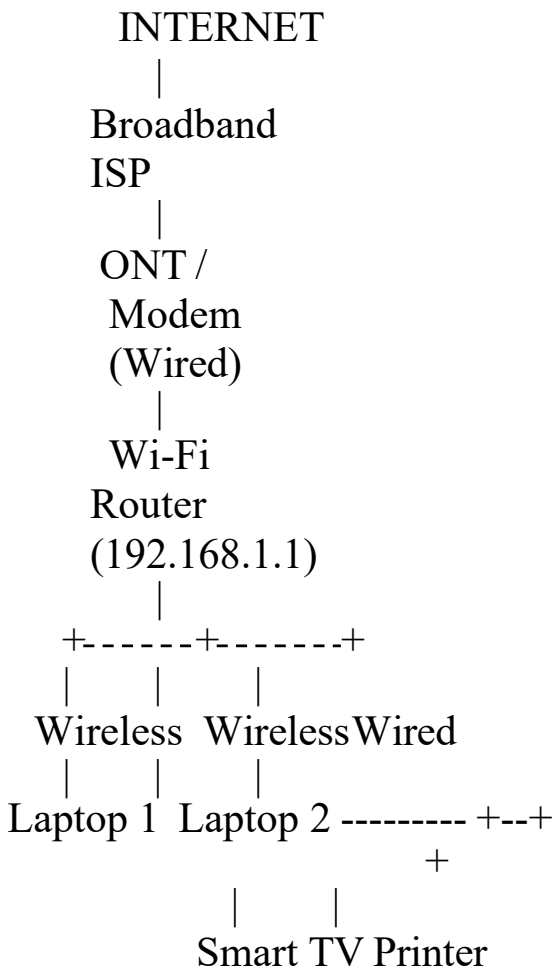
1

Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

A SOHO network connects multiple devices to the internet using a Wi-Fi router. The broadband connection provides internet access, and the router distributes the connection to all devices.

Network Diagram:



Connection Types:

1. Broadband to modem: Wired (Optical Fibre).
2. Modem to Wi-Fi router: Wired (Ethernet).
3. Router to Laptop 1: Wireless (Wi-Fi).
4. Router to Laptop 2: Wireless (Wi-Fi).
5. Router to Smart TV: Wired (Ethernet).

6. Router to Printer: Wired (Ethernet).

This network allows all five devices to share the same internet connection.

2.

List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Sr. No.	Hardware Component	Suggested Brand/Model	Purpose
1	Wi-Fi Router	TP-Link Archer C6 AC1200	Connects all devices to the network.
2	Broadband Modem/ONT	ISP-provided ONT	Receives internet from the service provider.
3	Laptop 1	HP Laptop	Connects to the network wirelessly.
4	Laptop 2	Dell Laptop	Connects to the network wirelessly.
5	Smart TV	Samsung Smart TV	Provides internet access for streaming.
6	Network Printer	HP LaserJet Pro	Allows wireless network users to print.
7	Ethernet Cable	D-Link Cat6 Cable	Connects the modem, TV and printer.
8	UPS	APC 600VA UPS	Provides backup power during electricity failures.

Additional Components:

- RJ45 connectors (if making custom Ethernet cables).
- Power adapters for the router and ONT.
- Wi-Fi range extender (optional, if the Wi-Fi signal is weak).
- Network switch (optional, if additional wired devices are required).

3.

Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

The following steps can be used to configure the TP-Link Archer C6 Wi-Fi router.

Step 1: Connect the Router

- Connect the broadband modem/ONT to the WAN port of the router using an Ethernet cable.
- Switch on the router and wait for the power and internet indicators.

Step 2: Open the Router Admin Page

- Connect a laptop to the router.
- Open a web browser.
- Enter 192.168.1.1in the address bar.
- Enter the administrator login credentials.

Step 3: Change the Admin Password

- Open System Tools or Administration settings.
- Select Password or Administrator Settings.
- Change the default admin password to a strong, unique password.
- Save the changes.

Step 4: Create a Unique SSID

- Open Wireless Settings.
- Enter a unique Wi-Fi network name

(SSID). Example: SOHO_Home_Network

Step 5: Enable WPA2 Security

- Open Wireless Security settings.
- Select WPA2-PSK (Personal).
- Select AES encryption.
- Set a strong Wi-Fi password.
- Save the settings.

Step 6: Enable DHCP

- Open DHCP Server settings.
- Enable the DHCP server.
- Set the IP address range from 192.168.1.100to 192.168.1.199.
- Save the configuration.

Step 7: Connect All Devices

- Connect both laptops using Wi-Fi.
- Connect the smart TV and printer using Ethernet cables.
- Enter the Wi-Fi password wherever required.

Step 8: Test the Internet Connection

Open a browser on both laptops and visit a website to check whether the internet is working.

4.
Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

For this SOHO network, the router will use the IP address 192.168.1.1.

The printer and smart TV will have static IP addresses, while both laptops will receive IP addresses automatically through DHCP.

Network Configuration:

- Network Address: 192.168.1.0/24
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server: 8.8.8.8
- DHCP Range: 192.168.1.100 – 192.168.1.199

IP Address Assignment Table:

Device	MAC Address (Sample)	IP Address	Assignment Type
Wi-Fi Router	02:11:22:33:44:01	192.168.1.1	Static
Printer	02:11:22:33:44:02	192.168.1.10	Static
Smart TV	02:11:22:33:44:03	192.168.1.20	Static
Laptop 1	02:11:22:33:44:04	192.168.1.100	DHCP
Laptop 2	02:11:22:33:44:05	192.168.1.101	DHCP

Configuration Steps:

1. Open the router administration page.
2. Configure the printer IP as 192.168.1.10.
3. Configure the smart TV IP as 192.168.1.20.
4. Keep DHCP enabled on the router.
5. Set both laptops to obtain an IP address automatically.
6. Check the assigned IP addresses using the router's DHCP client

list or the
ipconfigcommand.

5.

Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

If one laptop cannot connect to the internet, I would check the following possible causes and solutions.

1. Incorrect Wi-Fi Password

Problem: The laptop may be using an incorrect Wi-Fi password.

Solution:

- Open Wi-Fi settings on the laptop.
- Forget the saved network.
- Select the correct SSID.
- Enter the correct Wi-Fi password.
- Try connecting again.

2. Incorrect IP Address Configuration

Problem: The laptop may not be receiving a valid IP address from the DHCP server.

Solution:

- Open Command Prompt.
- Type ipconfigand press Enter.
- Check the IP address and default gateway.
- If necessary, type ipconfig /release.
- Then type ipconfig /renew.
- Check whether the laptop receives a valid IP address.

3. Weak Wi-Fi Signal

Problem: The laptop may be too far away from the Wi-Fi router.

Solution:

- Move the laptop closer to the router.

- Check the Wi-Fi signal strength.
- Restart the router if necessary.
- Use a Wi-Fi extender if the signal is consistently weak.

4. Router or Internet Connection Problem

Problem: The router or broadband connection may not be working properly.

Solution:

- Check the router's power and internet indicators.
- Restart the router and modem.
- Check whether other devices can access the internet.
- Contact the internet service provider if the connection is still unavailable.

5. Network Driver Problem

Problem: The laptop's Wi-Fi driver may be outdated or corrupted.

Solution:

- Open Device Manager.
- Expand Network Adapters.
- Find the wireless network adapter.
- Update the driver.
- Restart the laptop and try connecting again.